

In-Class Exercise Sheet 7

The topic for this lecture is making data *tidy* for easier data analysis.

1. We discuss the article on “Tidy Data” by Hadley Wickham and fill in the blanks.

(a) The author quotes that “_____ % of data analysis is spent on the process of cleaning and preparing the data.”

(b) In tidy data,

- each type of _____ forms a table,
- each _____ forms a row,
- each _____ forms a column,
- each cell is an _____.

region	year	population	region	year	population	region	year	population
Toronto	2016	2235145	Toronto	2016	2235145	Toronto	2016	2235145
Vancouver	2016	1027613	Vancouver	2016	1027613	Vancouver	2016	1027613
Montreal	2016	1823281	Montreal	2016	1823281	Montreal	2016	1823281
Calgary	2016	544870	Calgary	2016	544870	Calgary	2016	544870
Ottawa	2016	571146	Ottawa	2016	571146	Ottawa	2016	571146
Winnipeg	2016	321484	Winnipeg	2016	321484	Winnipeg	2016	321484

Illustration according to the book “Data Science: A First Introduction” by Tiffany Timbers, Trevor Campbell, Melissa Lee.

2. Consider the example of student grades for two exams in a course.

(a) A compact way to represent data for visual digestion is the following representation:

exam	Rozz	Andrew	Susie
midterm	1.3	2.0	1.7
final	2.3	1.7	1.0

This representation of the data is not tidy. Why? Tidy the data.

(b) Tidy or untidy?

student	stu_number	exam	grade
Rozz	666	midterm	1.3
Andrew	1969	midterm	2.0
Susie	314	midterm	1.7
Rozz	666	final	2.3
Andrew	1969	final	1.7
Susie	314	final	1.0

(c) Tidy or untidy?

subject_id	choices	reaction_times
1	A,B,B	312.3 433.4 365.1
2	B,A,B	393.1 491.0 372.2
3	B,A,A	356.3 313.9 475.5
4	A,B,B	292.4 352.8 378.1

Based on examples by Michael Franke, see <https://michael-franke.github.io/intro-data-analysis/Chap-02-02-data-tidy-data.html>. The examples were slightly adapted.

3. Tidy or untidy?

Afghanistan			Brazil			China		
year	cases	population	year	cases	population	year	cases	population
1999	745	19987071	1999	37737	172006362	1999	212258	1272915272
2000	2666	20595360	2000	80488	174504898	2000	213766	1280428583

Example from the online course “Data analysis and visualization in R (IN2339)”, TU Munich.

4. Suppose there are two variables, sex (male or female) and handedness (right- or left-handed). Further suppose that 100 individuals are randomly sampled from a very large population as part of a study of sex differences in handedness.

A contingency table can be created to display the numbers of individuals who are male right-handed and left-handed, female right-handed and left-handed. Contingency tables provide a basic picture of the interrelation between two variables and can help find interactions between them.

Such a contingency table is shown below.

	Right-handed	Left-handed	Total
Male	43	9	52
Female	44	4	48
Total	87	13	100

The numbers of the males, females, and right- and left-handed individuals are called marginal totals. The grand total (the total number of individuals represented in the contingency table) is the number in the bottom right corner.

(a) Write a SQL statement to create a tidy table for the contingency table.

(b) How many male and female individuals were studied? Write a SQL query.

- (c) How many left- and right-handed individuals were studied? Write a SQL query.
- (d) Out of all males, what fraction are left-handed? Write a SQL query.
- (e) What are the odds that a male is left-handed? Using SQL, compute the fraction of how many have the property compared to how many do not.
- (f) The strength of an association can be measured by the *odds ratio*: It quantifies the strength of the association between two events. The odds ratio is defined as the ratio of the odds of event *A* taking place in the presence of *B*, and the odds of *A* in the absence of *B*.
Write a SQL query.

Example adapted from Exercise 3.46 from the book “SQL for Data Science” by Antonio Badia, and the Wikipedia page https://en.wikipedia.org/wiki/Contingency_table.

5. Assume the table below:

earthquakes			
magnitude	y2000	y2001	y2002
4.0–4.9	7425	7456	7489
5.0–5.9	1318	1299	1312
6.0+	165	174	160

- (a) Sketch the tidy version of this table.
- (b) Sketch the output of the following SQL query:

```
SELECT magnitude, year, y2000, y2001, y2002
FROM earthquakes, (VALUES (2000), (2001), (2002)) as temp(year);
```

- (c) Based on (b), complete the SQL statement below to create the tidy table.

```
CREATE TABLE earthquakesTidy AS
```

```
SELECT _____
_____
_____
_____
_____
_____
```

```
FROM earthquakes, (VALUES (2000), (2001), (2002)) as temp(year);
```

(d) Complete the SQL statement below to reverse the process.

SELECT magnitude,

FROM earthquakesTidy GROUP BY magnitude -- (required here, why?);

Based on the book “SQL for Data Science” by Antonio Badia.

6. We are discussing metadata and research workflows. Fill in the blanks.

- The process of exploring, cleaning and preparing data is _____.
- A _____ is a sequence of actions applied to the data or to the results of earlier actions and helps keep track of how datasets are created or modified.
- Most actions can be done in two modes: _____ (data is changed and original version is lost),
and _____ (a new dataset is created, original version is kept).
- Actions can further be classified as _____ (changes can be undone)
and _____ (changes cannot be undone).

From the book “SQL for Data Science” by Antonio Badia.

7. Consider the following SQL statement:

```
ALTER TABLE Customers DROP COLUMN phone;
```

Which statements are accurate?

- ☐ The action is non-destructive. ☐ The action is reversible. ☐ None of these options.

8. Consider the following SQL statement:

```
CREATE TABLE CleanCustomers AS  
SELECT id, TRIM(name) AS name, email  
FROM Customers;
```

Which statements are accurate?

- ☐ The action is non-destructive. ☐ The action is reversible. ☐ None of these options.

9. Give one example of an action that is destructive but still reversible.

10. Give one example of an action that is destructive and not reversible.